Bernát Gábor

+1-747-777-0357 | gaborjbernat@gmail.com | Los Angeles, USA
bernat.tech | github.com/gaborbernat | linkedin.com/in/gaborbernat

SUMMARY

Senior Software Engineer at Bloomberg working at principal scope on developer platforms for its 10,000+ engineers, including artifact hosting for 16 package ecosystems and secret scanning. Designed the artifact portal, the linting service and the data-provenance and quality pipelines, all adopted in production; applies LLMs to developer tooling, including a classifier that cut a secret-review queue more than 200-fold on a production run. On call in every team since 2017. Lead maintainer of virtualenv, tox, filelock, platformdirs and pipx (over 1.5 billion downloads a month), working with AI agents under human design and review: in 2026, 696 releases, 143 vulnerabilities fixed and an open issue backlog cut from 447 to 2. Python Software Foundation Fellow and Python Packaging Authority member.

EXPERIENCE

Bloomberg Apr 2025 – present
Senior Software Engineer – Developer Experience Los Angeles, USA (remote)

- **AI in developer tooling.** Cut the secret-scanning review queue more than 200-fold on a production run with an LLM classifier. Built usage tooling (a Claude Code status line, macOS and Windows tray apps, a reworked model-usage dashboard) that shows engineers their token use against quota across Claude, GPT and internal models, so they can choose which model to use for which task and when.
- **Artifact portal.** Designed the self-service portal for the Artifactory platform the team runs for 16 ecosystems, from pip and npm to Maven, Conan, Docker and Helm: search, wheel compatibility filtering, yank and restore, and install commands.
- **Change feed.** Built the webhook, Kafka and PostgreSQL feed behind the portal so pages no longer wait on Artifactory, and rewrote the catalog sync to write only what changed: 212 s to 13 s, with download counts for 29,089 packages updating again.
- **Secret scanning.** Took over the company's secret scanning and extended it to full git history, fitting a 52-day single-machine scan into a weekend window, and to generic secrets that match no known format. Co-designed remediation around separate queues for the person who leaked a secret, the repository owners and triage.
- **Code search.** Moved company code search to authenticated access, contributing signed-JWT, PKCE and multi-issuer support to the shared auth agent. Moved indexing to object storage with incremental updates per repository, ended 503 errors during reindexing, and cut indexer logs by 95%.
- **Linting.** Designed the company's linting service and its integration kit, and integrated 14 linters and formatters that run in hundreds of repositories.

Bloomberg Apr 2016 – Apr 2025
Software Engineer, Senior from 2019 – Data Technologies London; Los Angeles from 2022

- **Data validation.** Core developer, from 2016, of the rules engine where analysts, most of them not programmers, write Python checks on financial data before customers see it; 500+ internal repositories build on its base service.
- **Data provenance and quality.** Designed the data-provenance service and the move of the quality-metrics pipeline from Hadoop batch jobs to Argo and Spark over terabyte inputs, both adopted in production; cut ingest memory from 1.4 GB to 520 MB and merged four services' lineage schemas into one shared library.
- **Shared infrastructure.** Wrote the Jenkins library 40 projects across teams build on, rebuilt a reporting team's Jenkins as a high-availability cluster still running in 2026, cut the S3 poller's memory from 1.14 GB to 521 MB, and built the event gateway and schemas for Bloomberg Second Measure.

Bloomberg – Python Guild (volunteer) 2017 – Nov 2026
Conferences Chair until 2022, then Packaging Working Group lead

- Led the Packaging Working Group; drove PEP 751 lock-file adoption across the internal build, dependency-update and install tools, and kept the RPC client libraries about 80 services depend on current from Python 3.8 to 3.15.
- Replaced a 2019 single sign-on library with tested FastAPI and Flask integrations about ten apps use.
- Rebuilt the engineering conference site, made it accessible, and cut its event sync from 4,500 queries to 600.

Gravity Research & Development 2011 – 2016
Software and Integration Engineer Budapest, Hungary
Java SaaS recommendation engine serving hundreds of millions of recommendations a day to dailymotion.com and allegro.pl.

OPEN SOURCE

Maintainer of 35+ projects 2015 – present
bernat.tech/oss

- **Core Python tooling.** Lead maintainer of platformdirs, filelock, virtualenv, tox and pipx, three of them in PyPI's top 150, and primary author of most of them; co-maintains build and brought it to 100% test coverage.
- **Rewrites and features.** Led the virtualenv 20 (2020) and tox 4 (2022) rewrites, writing over 90% and 80% of their commits. Added PEP 751 lock files and PEP 723 inline script metadata to tox, and replaced pipdeptree's engine with an agent-written Rust extension verified to produce identical output: 4.6× faster end to end, fully covered, no unsafe code.
- **New projects.** Forked appdirs into platformdirs (2021), now 22× appdirs' downloads and vendored by pip. Created pyproject-fmt, tox-uv, turbohtml (3–22× faster than the standard library) and Quotaline.
- **AI-assisted maintenance.** Agents write about 90% of the shipped code; design and merge decisions stay with the maintainer. Larger changes start from a reviewed plan, review depth scales with risk, and every change must pass tests with full diff coverage and a second-agent review before a manual one. Wrote virtualenv's AI contribution policy. In 2026 the open issue backlog across 12 projects went from 447 to 2 while new reports doubled, with 2 reverts across 5,760 merged pull requests.
- **Supply-chain security.** Selected for the GitHub Secure Open Source Fund (September 2026). Fixed and disclosed 143 vulnerabilities across seven long-standing projects in 2026, each with a patched release, many from triaging Anthropic's OSS Scanner reports. Hardened virtualenv's release pipeline (Trusted Publishing, linted workflows, build and SBOM attestations, reproducible wheels); each led project holds the OpenSSF Best Practices badge, and two run on OSS-Fuzz.
- **Upstream.** 16 merged CPython pull requests, including speedups to `logging.getLogger()` and `json.dumps()`; authored PEP 662 on editable installs (2021; PEP 660 was chosen instead); upstreamed search API fixes and features to OpenGrok and scanning-engine fixes to TruffleHog.

SPEAKING AND WRITING

21 talks, tutorials and workshops since 2018, mainly at PyCon US, EuroPython, PyTexas, PyBay, PyBeach and PyLondinium (recordings). Recent: *How to be a Better Software Engineer in the World of AI Agents* (PyBay 2026), *Becoming a Better Python Developer with AI*, a workshop on planning, testing and reviewing AI-written code (PyTexas 2026), and *Securing Your Python Supply Chain* (PyBeach 2026). Blog: bernat.tech/posts.

EDUCATION

Budapest University of Technology and Economics 2011 – 2013
Master's degree, Engineering Information Technology (grade 4.57/5) Hungary
Sapientia – Hungarian University of Transylvania 2007 – 2011
Bachelor's degree, Computer Science (grade 9.70/10) Romania

SKILLS

Languages *Python* (primary), Go, Rust, TypeScript, Swift, Kotlin, C
Tools Claude Code, Codex, Artifactory, Docker, Kubernetes, Kafka, Spark, Argo, PostgreSQL, Jenkins